UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

FACULTAD DE CIENCIAS FÍSICAS

Teoría General de la Relatividad

Práctica Calificada 03
(Segundo semestre 2025)

Indicaciones Generales:
Materiales a utilizar: libros y apuntes.
Puntaje total: 20 puntos.
Las soluciones deben ser presentadas en el orden establecido de la práctica.
Fecha de entrega: **16/11/2025**.
Enviar la práctica resuelta al siguiente correo: **richard.avalos@unmsm.edu.pe**

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1. Reducir las ecuaciones de Maxwell a dos ecuaciones tensoriales.

Métrica = $(+,-,-,-)$

$A_\mu = (A^0, A^1, A^2, A^3)$

Cuatripotencial $A_\mu = (\phi, -A)$

$A_\mu = (A_t, -A_x, -A_y, -A_z)$

Potencial escalar V/c

Cuadrigradiente $\partial^\mu = (\partial^0, \partial^1, \partial^2, \partial^3)$

$= \left(-\frac{1}{c}\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$

Electromagnetismo $\vec{E} = -\nabla V - \frac{\partial}{\partial t}\vec{A}$

$$\begin{cases} \vec{E}_x = -\frac{\partial V}{\partial x} - \frac{\partial}{\partial t}A_x \\ \vec{E}_y = -\frac{\partial V}{\partial y} - \frac{\partial}{\partial t}A_y \\ \vec{E}_z = -\frac{\partial V}{\partial z} - \frac{\partial}{\partial t}A_z \end{cases} \rightarrow \begin{cases} -c\partial^1 A^0 + c\partial^0 A^1 = F^{01} \\ -c\partial^2 A^0 + c\partial^0 A^2 = F^{02} \\ -c\partial^3 A^0 + c\partial^0 A^3 = F^{03} \end{cases}$$

$\vec{B} = \nabla \times \vec{A}$

$$\begin{cases} B_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \\ B_y = \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \\ B_z = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \end{cases} \rightarrow \begin{cases} \partial^2 A^3 - \partial^3 A^2 = F^{23} \\ \partial^3 A^1 - \partial^1 A^3 = -F^{13} \\ \partial^1 A^2 - \partial^2 A^1 = F^{12} \end{cases}$$

$\partial^0 A^0 - \partial^0 A^0 = F^{00} = 0$

Tensor electromagnético

$$F^{\mu\nu} = \begin{pmatrix} F^{00} & F^{01} & F^{02} & F^{03} \\ F^{10} & F^{11} & F^{12} & F^{13} \\ F^{20} & F^{21} & F^{22} & F^{23} \\ F^{30} & F^{31} & F^{32} & F^{33} \end{pmatrix} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}$$

$$\nabla \cdot B = 0 \qquad \nabla \cdot E = \frac{\rho}{\epsilon_0}$$

$$\nabla \times E + \frac{\partial}{\partial t} B = 0 \qquad \nabla \times B - \frac{1}{c^2}\frac{\partial}{\partial t} E = \mu_0 J$$

4 vector corriente $J^\nu = (c\rho, J_x, J_y, J_z)$

$\partial_1 F^{10} = -\partial^1 F^{10} = -\partial/\partial x\, E_x$

$\Rightarrow$ con $\nu = 0$

$$\partial_\mu F^{\mu 0} = \partial_0 F^{00} + \partial_1 F^{10} + \partial_2 F^{20} + \partial_3 F^{30}$$

($\partial_0 F^{00}$ tachado: $F^{00} = 0$)

$$\partial_i F^{i0} = \frac{\nabla \cdot E}{c}$$

$$J^0 = c\rho$$

} recuperamos la ec. de Maxwell

$$\nabla \cdot E = \frac{\rho}{\epsilon_0}$$

$$c \cdot \partial_\mu F^{\mu 0} = \frac{J^0}{c\epsilon_0}$$

$$\mu_0 = \frac{1}{\epsilon_0 c^2}$$

$$\boxed{\partial_\mu F^{\mu 0} = \mu_0 J^0}$$

$\Rightarrow$ con $\nu \neq 0$

$$\partial_\mu F^{\mu i}$$

$F^{ij} = \varepsilon^{ijk} B_k$

$$\partial_0 F^{0i} + \partial_j F^{ji}$$

$F^{213} = \varepsilon_{123} B_3 = -B_3$

$$-\frac{1}{c^2}\frac{\partial E^{01}}{\partial t} + \partial_1 F^{11} + \partial_2 F^{21} + \partial_3 F^{31}$$

($\partial_1 F^{11}$ tachado: $F^{11} = 0$; $F^{31} = \varepsilon$)

$$-\partial_2(B_3) + \partial_3(B_2)$$

$$-\frac{1}{c^2}\frac{\partial E^{0i}}{\partial t}$$

$$\partial^2(B_3) - \partial^3(B_2$$

$\partial^2 = \partial/\partial y$

$$-\frac{1}{c^2}\partial/\partial t\, E \qquad = \varepsilon^{ijk}\, \partial/\partial x_j\, B_k$$

$$\nabla \times B$$

$J^i = J_x, J_y, J_z$

recuperamos la ec. de Maxwell

$$\nabla \times B - \frac{1}{c^2}\frac{\partial}{\partial t} E = \mu_0 J$$

$$\boxed{\partial_j F^{ji} + \partial_0 F^{0i} = \mu_0 J^i}$$

Entonces con $\nu = (0, 3)$,
las 2 ecuaciones se reducen a:

* Dinámica del campo

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$$

Con $F^{\mu\nu}$ relacionados solo a B:

$$\partial^3 F^{21} + \partial^2 F^{13} + \partial^1 F^{32} = 0$$

$$\underbrace{\partial/\partial z(-B_z) + \partial/\partial y(-B_y) + \partial/\partial x(-B_x) = 0}$$

recuperamos la ec. de Maxwell

$$\nabla \cdot B = 0$$

y para $F^{\mu\nu}$ relacionados con índice 0.

$$\partial^0 F^{ij} + \partial^i F^{j0} + \partial_j F^{0i} = 0$$

$$\partial^0 \varepsilon^{ijk} B_k + \partial^i F^{j0} + \partial^j F^{0i} = 0$$

$$\cancel{\frac{1}{c}\frac{\partial}{\partial t}\varepsilon^{ijk} B_k} - \partial_i F^{j0} - \partial_j F^{0i} = 0$$

$$\cancel{-\partial_i(-E_j)} \cancel{-\partial_j(E_i)}$$

$$\frac{1}{c}\frac{\partial}{\partial t}\varepsilon^{ijk} B_k + \frac{1}{c}(-\partial_i E_j + \partial_j E_i) = 0 \qquad \partial_i E_j - \partial_j E_i = (\nabla \times E)_k \varepsilon_{ijk}$$

bajamos

$$\underbrace{-\varepsilon_{ijk}\frac{\partial}{\partial t}B_k - (\nabla \times E)_k \varepsilon_{ijk} = 0}$$

recuperamos la ec. de Maxwell

Entonces con $\nu = (0,3)$,
las 2 ecuaciones se reducen a:

$$\partial/\partial t\, B_k + \nabla \times E = 0$$

y Identidad de Bianchi

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0$$

2. Demuestre que las componentes del campo magnético están dadas por $B^k = -\frac{1}{2}\epsilon^{klm}F_{lm}$.

$$F^{\mu\nu} = \begin{pmatrix} F^{00} & F^{01} & F^{02} & F^{03} \\ F^{10} & F^{11} & F^{12} & F^{13} \\ F^{20} & F^{21} & F^{22} & F^{23} \\ F^{30} & F^{31} & F^{32} & F^{33} \end{pmatrix} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}$$

$B^x = 1/2\,\epsilon^{xlm}F_{lm}$:

$$B_x = 1/2\,\epsilon_{xlm}F^{lm} = 1/2\,\epsilon_{123}F^{23} + 1/2\,\epsilon_{132}F^{32} = 1/2\,B_x + 1/2\,(-1)(-B_x) = B_x$$

$$B_y = 1/2\,\epsilon_{ylm}F^{lm} = 1/2\,\epsilon_{213}F^{13} + 1/2\,\epsilon_{231}F^{31} = 1/2\,B_y + 1/2\,(-1)(-B_y) = B_y$$

$$B_z = 1/2\,\epsilon_{zlm}F^{lm} = 1/2\,\epsilon_{312}F^{12} + 1/2\,\epsilon_{321}F^{21} = 1/2\,B_z + 1/2\,(-1)(-B_z) = B_z$$

3. Derive la fuerza de Lorentz en forma tensorial.

$$\frac{dp^\mu}{d\tau} = q F^{\mu\nu} u_\nu$$

$p^\mu = m u^\mu$ → Cuadrimomento

$u^\nu = \gamma(c, -v)$ → Cuadrivector Velocidad

Componente temporal: $\mu = 0$

$$\frac{dp^0}{d\tau} = q F^{0\nu} u_\nu$$

$$\frac{d}{d\tau}(mc\gamma) = q \frac{E_i}{c} u_i$$

$\frac{d}{d\tau} = \gamma \frac{d}{dt}$

$$\gamma/c \frac{dE}{dt} = q E_i/c \, (\gamma v_i)$$

$$\boxed{\frac{dE}{dt} = q(E \cdot v)}$$

Componente espacial: $\mu = i$

$$\frac{dp^i}{d\tau} = q F^{i\nu} u_\nu$$

$u_\nu = (\gamma c, -\gamma \mathbf{v})$

$F^{io} = E_i/c$

$F^{ij} = -\epsilon_{ijk} B_k$

$$\frac{dp^i}{d\tau} = q(E_i/c)(\gamma c) + q(-\epsilon_{ijk} B_k)(-\gamma v_j)$$

$$\gamma \frac{dp^i}{dt} = q\gamma E_i + q\, \epsilon_{ijk} B_k \gamma v_j$$

$$\gamma \frac{dp^i}{dt} = q\gamma \left(E_i + (\mathbf{v} \times \mathbf{B})_i \right)$$

$$\boxed{\frac{d\mathbf{P}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})}$$

4. Utilizando la transformación de Lorentz

$$t' = \gamma(t - \beta x) \qquad x' = \gamma(x - \beta t)$$
$$y' = y \qquad z' = z$$

y la expresión usual del tensor $F_{\mu\nu}$ en función de las componentes del campo eléctrico y el campo magnético:

a) Encuentre explícitamente todas las componentes del tensor

$$F'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}F_{\alpha\beta}.$$

b) Utilizando el resultado anterior, escriba las leyes de transformación de todas las componentes del campo eléctrico y del campo magnético.

c) Demuestre que $B^2 - E^2$ es invariante de Lorentz.

d) Demuestre que $\bar{E} \cdot \bar{B}$ es invariante de Lorentz.

a)

$$F_{\mu\nu} = \begin{pmatrix} F_{00} & F_{01} & F_{02} & F_{03} \\ F_{10} & F_{11} & F_{12} & F_{13} \\ F_{20} & F_{21} & F_{22} & F_{23} \\ F_{30} & F_{31} & F_{32} & F_{33} \end{pmatrix} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

$$t = \gamma(t' + \beta x') \qquad x = \gamma(x' + \beta t')$$
$$y' = y \qquad z' = z$$

$$\Lambda_{\alpha\mu} = \frac{\partial x^\alpha}{\partial x'^\mu} \begin{cases} \frac{\partial x_0}{\partial x'_0} = \frac{dt}{dt'} = \gamma, \quad \frac{\partial x_0}{\partial x'_1} = \frac{dt}{dx'} = \gamma\beta, \quad \frac{\partial x_0}{\partial x'_2} = \frac{dt}{dy'} = 0, \quad \frac{\partial x_0}{\partial x'_3} = \frac{dt}{dz'} = 0 \\ \frac{\partial x_1}{\partial x'_0} = \frac{dt}{dx'} = \gamma\beta \quad \cdots \end{cases}$$

$$\Lambda_{\alpha\mu} = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$F'_{\mu\nu} = \Lambda_{\alpha\mu} F_{\alpha\beta} \Lambda_{\beta\nu}$$

$$F'_{\mu\nu} = \Lambda_{\alpha\mu} F_{\alpha\beta} \Lambda_{\mu\beta} = \Lambda \cdot (F \cdot \Lambda^T) = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix} \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\frac{v^2}{c^2 - v^2} - \frac{c^2}{c^2 - v^2} = \frac{v^2 - c^2}{c^2 - v^2} = -1$$

$$\frac{1}{1 - \frac{v^2}{c^2}} \frac{v^2}{c^2} - \frac{1}{1 - \frac{v^2}{c^2}} =$$

$$E_x/c\,(\gamma^2\beta^2 - \gamma^2) = -E_x/c$$

$$= \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -E_x/c\cdot\gamma\beta & -E_x/c\cdot\gamma & -E_y/c & -E_z/c \\ E_x/c\cdot\gamma & E_x/c\cdot\gamma\beta & -B_z & B_y \\ E_y/c\cdot\gamma + \gamma\beta B_z & E_y/c\,\gamma\beta + B_z\gamma & 0 & -B_x \\ E_z/c\cdot\gamma - \gamma\beta B_y & E_z/c\,\gamma\beta - \gamma\beta y & B_x & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -E_x/c & -E_y/c\cdot\gamma - B_z\cdot\gamma\beta & -E_z/c\,\gamma + B_y\gamma\beta \\ \cancel{-E_x/c\cdot\gamma^2\beta + E_x/c\cdot\gamma^2\beta} & \cancel{-E_x/c\cdot\gamma^2 + E_x/c\,\gamma^2\beta^2} & -E_y/c\cdot\gamma\beta - B_z\gamma & -E_z/c\,\gamma\beta + B_y\gamma \\ \cancel{-E_x/c\,\gamma^2\beta^2 + E_x/c\cdot\gamma^2} \; E_x/c & 0 & 0 & -B_x \\ E_y/c\cdot\gamma + \gamma\beta B_z & E_y/c\,\gamma\beta + B_z\gamma & B_x & 0 \\ E_z/c\cdot\gamma - \gamma\beta B_y & E_z/c\,\gamma\beta - \gamma\beta y & & \end{pmatrix}$$

$$F'_{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c\cdot\gamma - B_z\cdot\gamma\beta & -E_z/c\,\gamma + B_y\gamma\beta \\ E_x/c & 0 & -E_y/c\cdot\gamma\beta - B_z\gamma & -E_z/c\,\gamma\beta + B_y\gamma \\ E_y/c\cdot\gamma + \gamma\beta B_z & E_y/c\,\gamma\beta + B_z\gamma & 0 & -B_x \\ E_z/c\cdot\gamma - \gamma\beta B_y & E_z/c\,\gamma\beta - \gamma\beta y & B_x & 0 \end{pmatrix}$$

b) $E'_{x/c} = E_{x/c}$, $E'_{y/c} = E_{y/c}\cdot\gamma + B_z\cdot\gamma\beta$, $E'_{z/c} = E_{z/c}\cdot\gamma - B_y\cdot\gamma\beta$

$B'_x = B_x$, $B'_y = -E_{z/c}\gamma\beta + B_y\cdot\gamma$, $B'_z = E_{y/c}\gamma\beta + B_z\cdot\gamma$

$$E'_x = E_x \qquad B'_x = B_x$$

$$E'_y = \gamma\left(E_y + B_z\cdot c\cdot\beta\right) \qquad B'_y = \gamma\left(B_y - E_{z/c}\cdot\beta\right)$$

$$E'_z = \gamma\left(E_z - B_y\cdot c\cdot\beta\right) \qquad B'_z = \gamma\left(B_z + E_{y/c}\cdot\beta\right)$$

c) $B'^2 - (E/c)'^2 =$

$$B'^2 = B'^2_x + B'^2_y + B'^2_z = B_x^2 + \gamma^2\left(B_y - E_z/c\cdot\beta\right)^2 + \gamma^2\left(B_z + E_y/c\cdot\beta\right)^2$$
$$= B_x^2 + \gamma^2\left(B_y^2 - 2\frac{E_z}{c}\cdot\beta B_y + \frac{E_z^2}{c^2}\cdot\beta^2\right) + \gamma^2\left(B_z^2 + 2\frac{E_y}{c}\cdot\beta\cdot B_z + \frac{E_y^2}{c^2}\cdot\beta^2\right)$$

$$\frac{E'^2}{c^2} = \frac{E'^2_x}{c^2} + \frac{E'^2_y}{c^2} + \frac{E'^2_z}{c^2} = \frac{E_x^2}{c^2} + \gamma^2_{/c^2}\left(E_y + B_z\cdot c\cdot\beta\right)^2 + \gamma^2_{/c^2}\left(E_z - B_y\cdot c\cdot\beta\right)^2$$
$$= \frac{E_x^2}{c^2} + \gamma^2_{/c^2}\left(E_y^2 + 2\cdot B_z\cdot c\cdot E_y\cdot\beta + B_z^2\cdot c^2\cdot\beta^2\right) + \gamma^2_{/c^2}\left(E_z^2 - 2E_z B_y\cdot c\cdot\beta + B_y^2\cdot c^2\beta^2\right)$$

$$B'^2 - (E/c)'^2 = B_x^2 - \frac{E_x^2}{c^2} + \gamma^2\Big[B_y^2 - 2\frac{E_z}{c}\beta B_y + \frac{E_z^2}{c^2}\cdot\beta^2 + B_z^2 + 2\frac{E_y}{c}\cdot\beta\cdot B_z + E_y^2/c^2\beta^2 -$$
$$- \frac{E_y^2}{c^2} - \frac{2B_z\cdot\cancel{c}\cdot E_y\cdot\beta}{\cancel{c^2}\,c} - \frac{B_z^2\cdot\cancel{c^2}\cdot\beta^2}{\cancel{c^2}} - \frac{E_z^2}{c^2} + \frac{2E_z B_y\cdot\cancel{c}\cdot\beta}{\cancel{c^2}\,c} - \frac{B_y^2\cdot\cancel{c^2}\cdot\beta^2}{\cancel{c^2}}\Big]$$

$$B'^2 - (E'/c)^2 = B_x^2 - \frac{E_x^2}{c^2} + \gamma^2\Big[B_y^2 - \cancel{2\frac{E_z}{c}\beta B_y} + \frac{E_z^2}{c^2}\cdot\beta^2 + B_z^2 + \cancel{2\frac{E_y}{c}\cdot\beta\cdot B_z} + E_y^2/c^2\beta^2 -$$
$$- \frac{E_y^2}{c^2} - \cancel{\frac{2B_z\cdot E_y\cdot\beta}{c}} - B_z^2\cdot\beta^2 - \frac{E_z^2}{c^2} + \cancel{\frac{2E_z B_y\cdot\beta}{c}} - B_y^2\cdot\beta^2\Big]$$

$$= B_x^2 - \frac{E_x^2}{c^2} + \gamma^2\Big(B_y^2 + \frac{E_z^2\cdot\beta^2}{c^2} + B_z^2 + \frac{E_y^2}{c^2}\cdot\beta^2 - \frac{E_y^2}{c^2} - B_z^2\cdot\beta^2 - \frac{E_z^2}{c^2} - B_y^2\cdot\beta^2\Big)$$

$$= B_x^2 - \frac{E_x^2}{c^2} + \cancel{\gamma^2(1-\beta^2)B_y^2}_{1} + \cancel{\gamma^2(1-\beta^2)B_z^2}_{1} + \cancel{\gamma^2(\beta^2-1)\frac{E_z^2}{c^2}}_{-1} + \cancel{\gamma^2(\beta^2-1)\frac{E_y^2}{c^2}}_{-1}$$

$$B'^2 - (E'/c)^2 = B_x^2 - E_x^2/c^2 + B_y^2 + B_z^2 - E_z^2/c^2 - E_y^2/c^2 = B^2 - (E/c)^2 \qquad \square$$

d) $\frac{E'}{c}\cdot B' = E'_{x/c}\cdot B'_x + E'_{y/c}\cdot B'_y + E'_{z/c}\,B'_z$

$$= \frac{E_x}{c}B_x + \gamma^2\frac{\left(E_y + B_z\cdot c\cdot\beta\right)\left(B_y - E_{z/c}\cdot\beta\right)}{c} + \gamma^2\frac{\left(E_z - B_y\cdot c\cdot\beta\right)\left(B_z + E_{y/c}\cdot\beta\right)}{c}$$

$$= \frac{E_x B_x}{c} + \gamma^2\Big[\Big(\frac{E_y}{c} + B_z\beta\Big)\Big(B_y - E_{z/c}\cdot\beta\Big) + \Big(\frac{E_z}{c} - B_y\beta\Big)\Big(B_z + E_{y/c}\cdot\beta\Big)\Big]$$

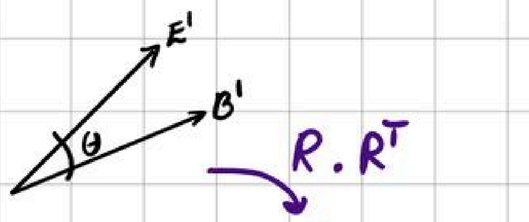
$$E_{y/c}B_y + \cancel{B_z\beta B_y}^{0} - \cancel{\frac{E_y}{c}\frac{E_z}{c}\cdot\beta}^{0} - B_z\beta^2\frac{E_z}{c} + \frac{E_z}{c}B_z - \cancel{B_y\beta B_z}^{0} + \cancel{\frac{E_z}{c}\frac{E_y}{c}\cdot\beta}^{0} - B_y\beta^2 E_{z/c}$$

$$\frac{E'}{c}\cdot B' = \frac{E_x B_x}{c} + \cancel{\gamma^2(1-\beta^2)}_{1}\Big(E_{y/c}B_y + \frac{E_z}{c}B_z\Big) = \frac{E_x B_x}{c} + \frac{E_y B_y}{c} + \frac{E_z B_z}{c} = \frac{E}{c}\cdot B \qquad \square$$

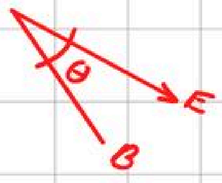
5. Los campos $\bar{E}$ y $\bar{B}$ de cierto campo electromagnético forman un ángulo θ ¿Cuál debe ser el valor del ángulo θ para que este sea invariante de Lorentz? Fundamente matemáticamente su respuesta.

Si $\frac{E'}{c} \cdot B' = \frac{E}{c} \cdot B$

$E' \cdot B' = E \cdot B$



$R \cdot R^T$



$|E'| \cdot |B'| \cos\theta = |E| \cdot |B| \cos\theta$

$\neq$

La única forma de que se cumpla la igualdad es que el $\cos\theta = 0$

Entonces solo se cumple para cuando $\theta = 90^\circ$

Lo que significa:

- Si $\vec{E} \cdot \vec{B} = 0$ en un sistema, entonces $\mathbf{E' \cdot B' = 0}$ en todos los sistemas, y $\theta = 90^\circ$ es Invariante.
- Si $\vec{E} \cdot \vec{B} \neq 0$ en un sistema, entonces $\cos\theta$ NO es Invariante por que $|E|$ y $|B|$ cambian de manera diferente bajo transformaciones de Lorentz, alterando el valor de $\cos\theta$.

6. Utilizando la regla de transformación de la conexión, demuestre explícitamente que las siguientes derivadas covariantes son tensores:

   a) $\nabla_\lambda T^{\mu\nu} = \partial_\lambda T^{\mu\nu} + \Gamma^\mu_{\lambda\sigma} T^{\sigma\nu} + \Gamma^\nu_{\lambda\sigma} T^{\mu\sigma}$

   b) $\nabla_\lambda T^\mu_{\ \nu} = \partial_\lambda T^\mu_{\ \nu} + \Gamma^\mu_{\lambda\sigma} T^\sigma_{\ \nu} - \Gamma^\sigma_{\lambda\nu} T^\mu_{\ \sigma}$

   c) $\nabla_\gamma T_{\mu\nu} = \partial_\lambda T_{\mu\nu} - \Gamma^\sigma_{\lambda\mu} T_{\sigma\nu} - \Gamma^\sigma_{\lambda\nu} T_{\mu\sigma}$

Para ser un tensor debe cumplir:

$$\nabla'_\lambda T'^{\mu\nu} = \frac{\partial x^\rho}{\partial x'^\lambda} \frac{\partial x'^\mu}{\partial x^\sigma} \frac{\partial x'^\nu}{\partial x^\tau} \nabla_\rho T^{\sigma\tau}$$

recordando:

$$\partial'_\lambda = \frac{\partial x^\rho}{\partial x'^\lambda} \partial_\rho \quad (*)$$

a) $$\nabla'_\lambda T'^{\mu\nu} = \underbrace{\partial'_\lambda T'^{\mu\nu}}_{①} + \underbrace{\Gamma'^{\mu}_{\lambda\sigma} T'^{\sigma\nu}}_{②} + \underbrace{\Gamma'^{\nu}_{\lambda\sigma} T'^{\mu\sigma}}_{③}$$

$$T'^{\mu\nu} = \frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta} T^{\alpha\beta}$$ → Relación entre derivadas $\partial'_\lambda = \frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho$:

①

→ $$\partial'_\lambda T'^{\mu\nu} = \frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta} T^{\alpha\beta}\right)$$

$$\partial'_\lambda T'^{\mu\nu} = \frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\alpha}\right)\frac{\partial x'^\nu}{\partial x^\beta} T^{\alpha\beta} + \frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\alpha}\partial_\rho\left(\frac{\partial x'^\nu}{\partial x^\beta}\right) T^{\alpha\beta} + \frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta}\partial_\rho\left(T^{\alpha\beta}\right)$$

→ Regla de transformación de la conexión:

$$\Gamma'^{\mu}_{\lambda\sigma} = \frac{\partial x'^\mu}{\partial x^\ell}\frac{\partial x^m}{\partial x'^\lambda}\frac{\partial x^n}{\partial x'^\sigma}\Gamma^\ell_{mn} + \frac{\partial x'^\mu}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\sigma}$$

② → $$\Gamma'^{\mu}_{\lambda\sigma} T'^{\sigma\nu} = \left(\frac{\partial x'^\mu}{\partial x^\ell}\frac{\partial x^m}{\partial x'^\lambda}\frac{\partial x^n}{\partial x'^\sigma}\Gamma^\ell_{mn} + \frac{\partial x'^\mu}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\sigma}\right)\left(\frac{\partial x'^\nu}{\partial x^\kappa}\frac{\partial x'^\sigma}{\partial x^\epsilon} T^{\epsilon\kappa}\right)$$

$$\Gamma'^{\mu}_{\lambda\sigma} T'^{\sigma\nu} = \frac{\partial x'^\mu}{\partial x^\ell}\frac{\partial x^m}{\partial x'^\lambda}\underbrace{\frac{\partial x^n}{\partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\epsilon}}_{\delta^n_\epsilon}\frac{\partial x'^\nu}{\partial x^\kappa}\Gamma^\ell_{mn} T^{\epsilon\kappa} + \frac{\partial x'^\mu}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\epsilon}\frac{\partial x'^\nu}{\partial x^\kappa} T^{\epsilon\kappa}$$

$$\Gamma'^{\mu}_{\lambda\sigma} T'^{\sigma\nu} = \frac{\partial x'^\mu}{\partial x^\ell}\frac{\partial x^m}{\partial x'^\lambda}\frac{\partial x'^\nu}{\partial x^\kappa}\Gamma^\ell_{mn} T^{n\kappa} + \frac{\partial x'^\mu}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\epsilon}\frac{\partial x'^\nu}{\partial x^\kappa} T^{\epsilon\kappa}$$

③ → $$\Gamma'^{\nu}_{\lambda\sigma} T'^{\mu\sigma} = \left(\frac{\partial x'^\nu}{\partial x^\ell}\frac{\partial x^m}{\partial x'^\lambda}\frac{\partial x^n}{\partial x'^\sigma}\Gamma^\ell_{mn} + \frac{\partial x'^\nu}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\sigma}\right)\left(\frac{\partial x'^\sigma}{\partial x^\epsilon}\frac{\partial x'^\mu}{\partial x^\kappa} T^{\kappa\epsilon}\right)$$

$$\Gamma'^{\nu}_{\lambda\sigma} T'^{\mu\sigma} = \frac{\partial x'^\nu}{\partial x^\ell}\frac{\partial x^m}{\partial x'^\lambda}\underbrace{\frac{\partial x^n}{\partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\epsilon}}_{\delta^n_\epsilon}\frac{\partial x'^\mu}{\partial x^\kappa}\Gamma^\ell_{mn} T^{\kappa\epsilon} + \frac{\partial x'^\nu}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\epsilon}\frac{\partial x'^\mu}{\partial x^\kappa} T^{\kappa\epsilon}$$

$$\Gamma'^{\nu}_{\lambda\sigma} T'^{\mu\sigma} = \frac{\partial x'^\nu}{\partial x^\ell}\frac{\partial x^m}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\kappa}\Gamma^\ell_{mn} T^{\kappa n} + \frac{\partial x'^\nu}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\epsilon}\frac{\partial x'^\mu}{\partial x^\kappa} T^{\kappa\epsilon}$$

Sumando ①, ② y ③ : , con ② $\ell=\alpha$, $m=\rho$, $k=\beta$ $\Gamma^\alpha_{\rho n} T^{n\beta}$, con ③ $\ell=\beta$, $m=\rho$, $k=\alpha$ $\Gamma^\beta_{\rho n} T^{\alpha n}$

$$\nabla'_\lambda T'^{\mu\nu} = \partial'_\lambda T'^{\mu\nu} + \Gamma'^{\mu}_{\lambda\sigma} T'^{\sigma\nu} + \Gamma'^{\nu}_{\lambda\sigma} T'^{\mu\sigma}$$

$$\frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta}\nabla_\rho T^{\alpha\beta} = \frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta}\left[\partial_\rho\left(T^{\alpha\beta}\right) + \Gamma^\alpha_{\rho n} T^{n\beta} + \Gamma^\beta_{\rho n} T^{\alpha n}\right]$$

Términos que sobran (∂^{bc} ↓ $= 0$):

$$\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\alpha}\right)\frac{\partial x'^\nu}{\partial x^\beta} T^{\alpha\beta} + \frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\alpha}\partial_\rho\left(\frac{\partial x'^\nu}{\partial x^\beta}\right) T^{\alpha\beta} + \frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial^2 x^\alpha}{\partial x'^\lambda \partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\epsilon}\frac{\partial x'^\nu}{\partial x^\beta} T^{\epsilon\beta} + \frac{\partial x'^\nu}{\partial x^\beta}\frac{\partial^2 x^\beta}{\partial x'^\lambda \partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\epsilon}\frac{\partial x'^\mu}{\partial x^\alpha} T^{\alpha\epsilon} = 0$$

con $\epsilon=\alpha$ con $\epsilon=\beta$

$$\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\alpha}\right)\frac{\partial x'^\nu}{\partial x^\beta}T^{\alpha\beta}+\frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\alpha}\partial_\rho\left(\frac{\partial x'^\nu}{\partial x^\beta}\right)T^{\alpha\beta}+\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial^2 x^\alpha}{\partial x'^\lambda\partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta}T^{\alpha\beta}+\frac{\partial x'^\nu}{\partial x^\beta}\frac{\partial^2 x^\beta}{\partial x'^\lambda\partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\beta}\frac{\partial x'^\mu}{\partial x^\alpha}T^{\alpha\beta}=0$$

I) ____ II) ____

I) $\quad \frac{\partial x'^\sigma}{\partial x^\alpha}\frac{\partial x'^\mu}{\partial x^\beta}\left(\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial^2 x^\alpha}{\partial x'^\lambda\partial x'^\sigma}\right)$

$$\frac{\partial}{\partial x'^\lambda}\left(\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\alpha}{\partial x'^\sigma}\right)=\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\alpha}\frac{\partial x^\alpha}{\partial x'^\sigma}+\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial^2 x^\alpha}{\partial x'^\lambda\partial x'^\sigma}$$

$$\frac{\partial}{\partial x'^\lambda}\left(\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\alpha}{\partial x'^\sigma}\right)=0$$

$$\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial^2 x^\alpha}{\partial x'^\lambda\partial x'^\sigma}=-\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\alpha}\frac{\partial x^\alpha}{\partial x'^\sigma}$$

$$-\frac{\partial x'^\sigma}{\partial x^\alpha}\frac{\partial x'^\mu}{\partial x^\beta}\left(\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\alpha}\frac{\partial x^\alpha}{\partial x'^\sigma}\right)=-\frac{\partial x'^\mu}{\partial x^\beta}\left(\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\alpha}\right)$$

usando (*)

$$-\frac{\partial x'^\mu}{\partial x^\beta}\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\alpha}\right)$$

II) $\quad \frac{\partial x'^\sigma}{\partial x^\beta}\frac{\partial x'^\mu}{\partial x^\alpha}\left(\frac{\partial x'^\mu}{\partial x^\beta}\frac{\partial^2 x^\beta}{\partial x'^\lambda\partial x'^\sigma}\right)$

$$\frac{\partial}{\partial x'^\lambda}\left(\frac{\partial x'^\mu}{\partial x^\beta}\frac{\partial x^\beta}{\partial x'^\sigma}\right)=\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\beta}\frac{\partial x^\beta}{\partial x'^\sigma}+\frac{\partial x'^\mu}{\partial x^\beta}\frac{\partial^2 x^\beta}{\partial x'^\lambda\partial x'^\sigma}$$

$$\frac{\partial}{\partial x'^\lambda}\left(\delta^\mu_\sigma\right)=0$$

$$\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\beta}\frac{\partial x^\beta}{\partial x'^\sigma}=-\frac{\partial x'^\mu}{\partial x^\beta}\frac{\partial^2 x^\beta}{\partial x'^\lambda\partial x'^\sigma}$$

$$-\frac{\partial x'^\sigma}{\partial x^\beta}\frac{\partial x'^\mu}{\partial x^\alpha}\left(\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\beta}\frac{\partial x^\beta}{\partial x'^\sigma}\right)=-\frac{\partial x'^\mu}{\partial x^\alpha}\left(\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\beta}\right)$$

usando (*)

$$-\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\beta}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\beta}\right)$$

$$\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\alpha}\right)\frac{\partial x'^\nu}{\partial x^\beta}T^{\alpha\beta}+\frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\alpha}\partial_\rho\left(\frac{\partial x'^\nu}{\partial x^\beta}\right)T^{\alpha\beta}-\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\alpha}\right)\frac{\partial x'^\nu}{\partial x^\beta}T^{\alpha\beta}-\frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\alpha}\partial_\rho\left(\frac{\partial x'^\nu}{\partial x^\beta}\right)T^{\alpha\beta}=0$$

(I) (II)

$$\square$$

$$\nabla_\rho T^{\alpha\beta}$$

$$\nabla'_\lambda T'^{\mu\nu}=\partial'_\lambda T'^{\mu\nu}+\Gamma'^\mu_{\lambda\sigma}T'^{\sigma\nu}+\Gamma'^\nu_{\lambda\sigma}T'^{\mu\sigma}=\frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\nu}{\partial x^\beta}\left[\partial_\rho\left(T^{\alpha\beta}\right)+\Gamma^\alpha_{\rho n}T^{n\beta}+\Gamma^\beta_{\rho n}T^{\alpha n}\right]$$

b) $\nabla'_\lambda T'^{\mu}{}_{\nu} = \underbrace{\partial'_\lambda T'^{\mu}{}_{\nu}}_{①} + \underbrace{\Gamma'^{\mu}_{\lambda\sigma} T'^{\sigma}{}_{\nu}}_{②} - \underbrace{\Gamma'^{\sigma}_{\lambda\nu} T'^{\mu}{}_{\sigma}}_{③}$

$T'^{\mu}{}_{\nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} T^{\alpha}{}_{\beta}$ → Relación entre derivadas $\partial'_\lambda = \frac{\partial x^{\rho}}{\partial x'^{\lambda}} \partial_\rho$ ∴

① → $\partial'_\lambda T'^{\mu}{}_{\nu} = \frac{\partial x^{\rho}}{\partial x'^{\lambda}} \partial_\rho \left(\frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} T^{\alpha}{}_{\beta} \right)$

$\partial'_\lambda T'^{\mu}{}_{\nu} = \frac{\partial x^{\rho}}{\partial x'^{\lambda}} \partial_\rho \left(\frac{\partial x'^{\mu}}{\partial x^{\alpha}} \right) \frac{\partial x^{\beta}}{\partial x'^{\nu}} T^{\alpha}{}_{\beta} + \frac{\partial x^{\rho}}{\partial x'^{\lambda}} \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \partial_\rho \left(\frac{\partial x^{\beta}}{\partial x'^{\nu}} \right) T^{\alpha}{}_{\beta} + \frac{\partial x^{\rho}}{\partial x'^{\lambda}} \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} \partial_\rho \left(T^{\alpha}{}_{\beta} \right)$

→ Regla de transformación de la conexión:

$$\Gamma'^{\mu}_{\lambda\sigma} = \frac{\partial x'^{\mu}}{\partial x^{\ell}} \frac{\partial x^{m}}{\partial x'^{\lambda}} \frac{\partial x^{n}}{\partial x'^{\sigma}} \Gamma^{\ell}_{mn} + \frac{\partial x'^{\mu}}{\partial x^{\ell}} \frac{\partial^2 x^{\ell}}{\partial x'^{\lambda} \partial x'^{\sigma}}$$

② → $\Gamma'^{\mu}_{\lambda\sigma} T'^{\sigma}{}_{\nu} = \left(\frac{\partial x'^{\mu}}{\partial x^{\ell}} \frac{\partial x^{m}}{\partial x'^{\lambda}} \frac{\partial x^{n}}{\partial x'^{\sigma}} \Gamma^{\ell}_{mn} + \frac{\partial x'^{\mu}}{\partial x^{\ell}} \frac{\partial^2 x^{\ell}}{\partial x'^{\lambda} \partial x'^{\sigma}} \right) \left(\frac{\partial x^{\kappa}}{\partial x'^{\nu}} \frac{\partial x'^{\sigma}}{\partial x^{\epsilon}} T^{\epsilon}{}_{\kappa} \right)$

$\Gamma'^{\mu}_{\lambda\sigma} T'^{\sigma}{}_{\nu} = \frac{\partial x'^{\mu}}{\partial x^{\ell}} \frac{\partial x^{m}}{\partial x'^{\lambda}} \underbrace{\frac{\partial x^{n}}{\partial x'^{\sigma}} \frac{\partial x'^{\sigma}}{\partial x^{\epsilon}}}_{\delta^{n}_{\epsilon}} \frac{\partial x^{\kappa}}{\partial x'^{\nu}} \Gamma^{\ell}_{mn} T^{\epsilon}{}_{\kappa} + \frac{\partial x'^{\mu}}{\partial x^{\ell}} \frac{\partial^2 x^{\ell}}{\partial x'^{\lambda} \partial x'^{\sigma}} \frac{\partial x'^{\sigma}}{\partial x^{\epsilon}} \frac{\partial x^{\kappa}}{\partial x'^{\nu}} T^{\epsilon}{}_{\kappa}$

$\Gamma'^{\mu}_{\lambda\sigma} T'^{\sigma}{}_{\nu} = \frac{\partial x'^{\mu}}{\partial x^{\ell}} \frac{\partial x^{m}}{\partial x'^{\lambda}} \frac{\partial x^{\kappa}}{\partial x'^{\nu}} \Gamma^{\ell}_{mn} T^{n}{}_{\kappa} + \frac{\partial x'^{\mu}}{\partial x^{\ell}} \frac{\partial^2 x^{\ell}}{\partial x'^{\lambda} \partial x'^{\sigma}} \frac{\partial x'^{\sigma}}{\partial x^{\epsilon}} \frac{\partial x^{\kappa}}{\partial x'^{\nu}} T^{\epsilon}{}_{\kappa}$

③ → $\Gamma'^{\sigma}_{\lambda\nu} T'^{\mu}{}_{\sigma} = \left(\frac{\partial x'^{\sigma}}{\partial x^{\ell}} \frac{\partial x^{m}}{\partial x'^{\lambda}} \frac{\partial x^{n}}{\partial x'^{\nu}} \Gamma^{\ell}_{mn} + \frac{\partial x'^{\sigma}}{\partial x^{\ell}} \frac{\partial^2 x^{\ell}}{\partial x'^{\lambda} \partial x'^{\nu}} \right) \left(\frac{\partial x^{\epsilon}}{\partial x'^{\sigma}} \frac{\partial x'^{\mu}}{\partial x^{\kappa}} T^{\kappa}{}_{\epsilon} \right)$

$\Gamma'^{\sigma}_{\lambda\nu} T'^{\mu}{}_{\sigma} = \frac{\partial x^{n}}{\partial x'^{\nu}} \frac{\partial x^{m}}{\partial x'^{\lambda}} \underbrace{\frac{\partial x'^{\sigma}}{\partial x^{\ell}} \frac{\partial x^{\epsilon}}{\partial x'^{\sigma}}}_{\delta^{\epsilon}_{\ell} = -\delta^{\ell}_{\epsilon}} \frac{\partial x'^{\mu}}{\partial x^{\kappa}} \Gamma^{\ell}_{mn} T^{\kappa}{}_{\epsilon} + \frac{\partial x'^{\sigma}}{\partial x^{\ell}} \frac{\partial^2 x^{\ell}}{\partial x'^{\lambda} \partial x'^{\nu}} \frac{\partial x^{\epsilon}}{\partial x'^{\sigma}} \frac{\partial x'^{\mu}}{\partial x^{\kappa}} T^{\kappa}{}_{\epsilon}$

$\Gamma'^{\sigma}_{\lambda\nu} T'^{\mu}{}_{\sigma} = - \frac{\partial x^{n}}{\partial x'^{\nu}} \frac{\partial x^{m}}{\partial x'^{\lambda}} \frac{\partial x'^{\mu}}{\partial x^{\kappa}} \Gamma^{\ell}_{mn} T^{k}_{\ell} + \frac{\partial x'^{\sigma}}{\partial x^{\ell}} \frac{\partial^2 x^{\ell}}{\partial x'^{\lambda} \partial x'^{\nu}} \frac{\partial x^{\epsilon}}{\partial x'^{\sigma}} \frac{\partial x'^{\mu}}{\partial x^{\kappa}} T^{\kappa}{}_{\epsilon}$

Sumando ①, ② y ③: , con ② $\ell=\alpha$, $m=\rho$, $\kappa=\beta$ $\Gamma^{\alpha}_{\rho n} T^{n}{}_{\beta}$, con ③ $n=\beta$, $m=\rho$, $\kappa=\alpha$ $\Gamma^{\ell}_{\rho\beta} \Gamma^{\alpha}_{\ell}$

$$\nabla'_\lambda T'^{\mu}{}_{\nu} = \partial'_\lambda T'^{\mu}{}_{\nu} + \Gamma'^{\mu}_{\lambda\sigma} T'^{\sigma}{}_{\nu} - \Gamma'^{\sigma}_{\lambda\nu} T'^{\mu}{}_{\sigma}$$

$$\frac{\partial x^{\rho}}{\partial x'^{\lambda}} \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} \nabla_\rho T^{\alpha}{}_{\beta} = \frac{\partial x^{\rho}}{\partial x'^{\lambda}} \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} \left[\partial_\rho \left(T^{\alpha}{}_{\beta} \right) + \Gamma^{\alpha}_{\rho n} T^{n}{}_{\beta} - \Gamma^{\ell}_{\rho\beta} \Gamma^{\alpha}_{\ell} \right]$$

Términos que sobran ∴ → debe ser

$\frac{\partial x^{\rho}}{\partial x'^{\lambda}} \partial_\rho \left(\frac{\partial x'^{\mu}}{\partial x^{\alpha}} \right) \frac{\partial x^{\beta}}{\partial x'^{\nu}} T^{\alpha}{}_{\beta} + \frac{\partial x^{\rho}}{\partial x'^{\lambda}} \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \partial_\rho \left(\frac{\partial x^{\beta}}{\partial x'^{\nu}} \right) T^{\alpha}{}_{\beta} + \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\sigma}} \frac{\partial x^{\epsilon}}{\partial x'^{\nu}} \frac{\partial x'^{\sigma}}{\partial x^{\rho}} \frac{\partial^2 x^{\ell}}{\partial x'^{\lambda} \partial x'^{\nu}} T^{\epsilon}{}_{\beta} + \frac{\partial x'^{\sigma}}{\partial x^{\ell}} \frac{\partial^2 x^{\ell}}{\partial x'^{\lambda} \partial x'^{\nu}} \frac{\partial x^{\epsilon}}{\partial x'^{\sigma}} \frac{\partial x'^{\mu}}{\partial x^{\alpha}} T^{\alpha}{}_{\epsilon} = 0$

con $\epsilon=\alpha$ con $\epsilon=\beta$

$$\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\alpha}\right)\frac{\partial x^\beta}{\partial x'^\nu}T^\alpha{}_\beta+\frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x'^\mu}{\partial x^\alpha}\partial_\rho\left(\frac{\partial x^\beta}{\partial x'^\nu}\right)T^\alpha{}_\beta+\underbrace{\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial^2 x^\alpha}{\partial x'^\lambda\partial x'^\sigma}\frac{\partial x'^\sigma}{\partial x^\alpha}\frac{\partial x^\beta}{\partial x'^\nu}T^\alpha{}_\beta}_{\text{I)}}+\underbrace{\frac{\partial x'^\sigma}{\partial x^l}\frac{\partial^2 x^l}{\partial x'^\lambda\partial x'^\nu}\frac{\partial x^\beta}{\partial x'^\sigma}\frac{\partial x'^\mu}{\partial x^\alpha}T^\alpha_\beta}_{\text{II)}}=0$$

I) $\dfrac{\partial x'^\sigma}{\partial x^\alpha}\dfrac{\partial x^\beta}{\partial x'^\nu}\left(\dfrac{\partial^2 x^\alpha}{\partial x'^\lambda\partial x'^\sigma}\dfrac{\partial x'^\mu}{\partial x^\alpha}\right)$

$$\frac{\partial}{\partial x'^\lambda}\left(\frac{\partial x^\alpha}{\partial x'^\sigma}\frac{\partial x'^\mu}{\partial x^\alpha}\right)=\frac{\partial^2 x^\alpha}{\partial x'^\lambda\partial x'^\sigma}\frac{\partial x'^\mu}{\partial x^\alpha}+\frac{\partial x^\alpha}{\partial x'^\sigma}\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\alpha}$$

$\delta^\mu_\sigma=0$

$$\frac{\partial^2 x^\alpha}{\partial x'^\lambda\partial x'^\sigma}\frac{\partial x'^\mu}{\partial x^\alpha}=-\frac{\partial x^\alpha}{\partial x'^\sigma}\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\alpha}$$

$$\cancel{\frac{\partial x'^\sigma}{\partial x^\alpha}}\frac{\partial x^\beta}{\partial x'^\nu}\left(\cancel{\frac{\partial x^\alpha}{\partial x'^\sigma}}\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\alpha}\right)=-\frac{\partial x^\beta}{\partial x'^\nu}\left(\frac{\partial^2 x'^\mu}{\partial x'^\lambda\partial x^\alpha}\right)$$ usando (*)

$$-\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\alpha}\right)$$

II) $\underbrace{\dfrac{\partial x'^\sigma}{\partial x^l}}\dfrac{\partial^2 x^l}{\partial x'^\lambda\partial x'^\nu}\underbrace{\dfrac{\partial x^\beta}{\partial x'^\sigma}}\dfrac{\partial x'^\mu}{\partial x^\alpha}$

$\delta^\beta_l=-\delta^l_\beta$

$$-\frac{\partial^2 x^\beta}{\partial x'^\lambda\partial x'^\nu}\frac{\partial x'^\mu}{\partial x^\alpha}$$

$$-\frac{\partial^2 x'^\beta}{\partial x'^\lambda\partial x'^\nu}\frac{\partial x'^\mu}{\partial x^\alpha}$$ usando (*)

$$-\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\beta}{\partial x^\nu}\right)$$

$$\cancel{\frac{\partial x'^\rho}{\partial x^\rho}\partial_\rho\left(\frac{\partial x'^\mu}{\partial x^\alpha}\right)\frac{\partial x^\beta}{\partial x'^\nu}T^\alpha{}_\beta}+\cancel{\frac{\partial x'^\rho}{\partial x^\rho}\frac{\partial x'^\mu}{\partial x^\alpha}\partial_\rho\left(\frac{\partial x^\beta}{\partial x'^\nu}\right)T^\alpha{}_\beta}-\underbrace{\cancel{\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\rho}{\partial x^\beta}\partial_\rho\left(\frac{\partial x'^\nu}{\partial x^\mu}\right)T^\alpha{}_\beta}}_{(I)}-\underbrace{\cancel{\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x'^\rho}{\partial x^\beta}\partial_\rho\left(\frac{\partial x'^\beta}{\partial x^\nu}\right)T^\alpha_\beta}}_{(II)}=0$$

$\nabla_\rho T^\alpha_\beta$

$$\nabla'_\lambda T'^\mu{}_\nu=\partial'_\lambda T'^\mu{}_\nu+\Gamma'^\mu_{\lambda\sigma}T'^\sigma{}_\nu-\Gamma'^\sigma_{\lambda\nu}T'^\mu{}_\sigma=\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x^\beta}{\partial x'^\nu}\left[\partial_\rho\left(T^\alpha{}_\beta\right)+\Gamma^\alpha_{\rho\eta}T^\eta{}_\beta-\Gamma^l_{\rho\beta}\Gamma^\alpha_{l}\right]$$

c) $$\nabla'_\lambda T'_{\mu\nu} = \underbrace{\partial'_\lambda T'_{\mu\nu}}_{①} - \underbrace{\Gamma'^{\sigma}_{\lambda\mu} T'_{\sigma\nu}}_{②} - \underbrace{\Gamma'^{\sigma}_{\lambda\nu} T'_{\mu\sigma}}_{③}$$

$$T'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu} T_{\alpha\beta}$$ → Relación entre derivadas $\partial'_\lambda = \frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho$:

① → $$\partial'_\lambda T'^{\mu}{}_{\nu} = \frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu} T_{\alpha\beta}\right)$$

$$\partial'_\lambda T'^{\mu}{}_{\nu} = \frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x^\alpha}{\partial x'^\mu}\right)\frac{\partial x^\beta}{\partial x'^\nu} T_{\alpha\beta} + \frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x^\alpha}{\partial x'^\mu}\partial_\rho\left(\frac{\partial x^\beta}{\partial x'^\nu}\right) T_{\alpha\beta} + \frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\partial_\rho\left(T_{\alpha\beta}\right)$$

→ Regla de transformación de la conexión:

$$\Gamma'^{\sigma}_{\lambda\mu} = \frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial x^m}{\partial x'^\lambda}\frac{\partial x^n}{\partial x'^\mu}\Gamma^{\ell}_{mn} + \frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\mu}$$

② → $$\Gamma'^{\sigma}_{\lambda\mu} T'_{\nu\tau} = \left(\frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial x^m}{\partial x'^\lambda}\frac{\partial x^n}{\partial x'^\mu}\Gamma^{\ell}_{mn} + \frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\mu}\right)\left(\frac{\partial x^\kappa}{\partial x'^\nu}\frac{\partial x^\epsilon}{\partial x'^\tau} T_{\epsilon\kappa}\right)$$

$$\Gamma'^{\sigma}_{\lambda\mu} T'^{\mu}{}_{\tau} = \frac{\partial x^n}{\partial x'^\mu}\frac{\partial x^m}{\partial x'^\lambda}\underbrace{\frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial x^\epsilon}{\partial x'^\tau}}_{\delta^\epsilon_\ell = -\delta^\ell_\epsilon}\frac{\partial x^\kappa}{\partial x'^\nu}\Gamma^{\ell}_{mn} T_{\epsilon\kappa} + \frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\mu}\frac{\partial x^\kappa}{\partial x'^\nu}\frac{\partial x^\epsilon}{\partial x'^\tau} T_{\epsilon\kappa}$$

$$\Gamma'^{\sigma}_{\lambda\mu} T'^{\mu}{}_{\tau} = -\frac{\partial x^n}{\partial x'^\mu}\frac{\partial x^m}{\partial x'^\lambda}\frac{\partial x^\kappa}{\partial x'^\nu}\Gamma^{\ell}_{mn} T_{\ell\kappa} + \frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\mu}\frac{\partial x^\kappa}{\partial x'^\nu}\frac{\partial x^\epsilon}{\partial x'^\tau} T_{\epsilon\kappa}$$

③ → $$\Gamma'^{\sigma}_{\lambda\nu} T'^{\mu}{}_{\tau} = \left(\frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial x^m}{\partial x'^\lambda}\frac{\partial x^n}{\partial x'^\nu}\Gamma^{\ell}_{mn} + \frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\nu}\right)\left(\frac{\partial x^\kappa}{\partial x'^\mu}\frac{\partial x^\epsilon}{\partial x'^\tau} T_{\kappa\epsilon}\right)$$

$$\Gamma'^{\sigma}_{\lambda\nu} T'^{\mu}{}_{\tau} = \frac{\partial x^n}{\partial x'^\nu}\frac{\partial x^m}{\partial x'^\lambda}\underbrace{\frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial x^\epsilon}{\partial x'^\tau}}_{\delta^\epsilon_\ell = -\delta^\ell_\epsilon}\frac{\partial x^\kappa}{\partial x'^\mu}\Gamma^{\ell}_{mn} T_{\kappa\epsilon} + \frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\nu}\frac{\partial x^\kappa}{\partial x'^\mu}\frac{\partial x^\epsilon}{\partial x'^\tau} T_{\kappa\epsilon}$$

$$\Gamma'^{\sigma}_{\lambda\nu} T'^{\mu}{}_{\tau} = -\frac{\partial x^n}{\partial x'^\nu}\frac{\partial x^m}{\partial x'^\lambda}\frac{\partial x^\kappa}{\partial x'^\mu}\Gamma^{\ell}_{mn} T_{\kappa\ell} + \frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\nu}\frac{\partial x^\kappa}{\partial x'^\mu}\frac{\partial x^\epsilon}{\partial x'^\tau} T_{\kappa\epsilon}$$

Sumando ①, ② y ③ : , con ② $n=\alpha, k=\beta, m=\rho \Rightarrow \Gamma^{\ell}_{\rho\alpha} T_{\ell\beta}$ con ③ $n=\beta, m=\rho, k=\alpha \Rightarrow \Gamma^{\ell}_{\rho\beta} T_{\alpha\ell}$

$$\nabla'_\lambda T'_{\mu\nu} = \underbrace{\partial'_\lambda T'_{\mu\nu}}_{①} - \underbrace{\Gamma'^{\sigma}_{\lambda\mu} T'_{\sigma\nu}}_{②} - \underbrace{\Gamma'^{\sigma}_{\lambda\nu} T'_{\mu\sigma}}_{③}$$

$$\frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\nabla_\rho T_{\alpha\beta} = \frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\left[\partial_\rho\left(T_{\alpha\beta}\right) - \Gamma^{\ell}_{\rho\alpha} T_{\ell\beta} - \Gamma^{\ell}_{\rho\beta} T_{\alpha\ell}\right]$$

Términos que sobran ∂cbe →

$$\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x^\alpha}{\partial x'^\mu}\right)\frac{\partial x^\beta}{\partial x'^\nu} T_{\alpha\beta} + \frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x^\alpha}{\partial x'^\mu}\partial_\rho\left(\frac{\partial x^\beta}{\partial x'^\nu}\right) T_{\alpha\beta} + \frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\epsilon}{\partial x'^\tau} T_{\epsilon\beta} + \frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\nu}\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\epsilon}{\partial x'^\tau} T_{\alpha\epsilon} = 0$$

$\epsilon = \alpha = \ell$ $\qquad$ $\epsilon = \beta = \ell$

$$\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x^\alpha}{\partial x'^\mu}\right)\frac{\partial x^\beta}{\partial x'^\nu}T_{\alpha\beta} + \frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x^\alpha}{\partial x'^\mu}\partial_\rho\left(\frac{\partial x^\beta}{\partial x'^\nu}\right)T_{\alpha\beta} + \underbrace{\frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\alpha}{\partial x'^\sigma}T_{\alpha\beta}}_{\text{I)}} + \underbrace{\frac{\partial x'^\sigma}{\partial x^\ell}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\nu}\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\sigma}T_{\alpha\beta}}_{\text{II)}}$$

I)
$$\underbrace{\frac{\partial x'^\sigma}{\partial x^\ell}}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\underbrace{\frac{\partial x^\alpha}{\partial x'^\sigma}}$$

$$\delta^\alpha_\ell = -\delta^\ell_\alpha$$

$$-\frac{\partial^2 x^\alpha}{\partial x'^\lambda \partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}$$

$$-\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x^\alpha}{\partial x'^\mu}\right)\frac{\partial x^\beta}{\partial x'^\nu}$$

usando (*)

II)
$$\underbrace{\frac{\partial x'^\sigma}{\partial x^\ell}}\frac{\partial^2 x^\ell}{\partial x'^\lambda \partial x'^\nu}\underbrace{\frac{\partial x^\beta}{\partial x'^\sigma}}\frac{\partial x^\alpha}{\partial x'^\mu}$$

$$\delta^\beta_\ell = -\delta^\ell_\beta$$

$$-\frac{\partial^2 x^\beta}{\partial x'^\lambda \partial x'^\nu}\frac{\partial x^\alpha}{\partial x'^\mu}$$

$$-\frac{\partial^2 x'^\beta}{\partial x'^\lambda \partial x'^\nu}\frac{\partial x^\alpha}{\partial x'^\mu}$$

usando (*)

$$-\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x'^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\beta}{\partial x^\nu}\right)$$

$$\cancel{\frac{\partial x^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x^\alpha}{\partial x'^\mu}\right)\frac{\partial x^\beta}{\partial x'^\nu}T_{\alpha\beta}} + \cancel{\frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x^\alpha}{\partial x'^\mu}\partial_\rho\left(\frac{\partial x^\beta}{\partial x'^\nu}\right)T_{\alpha\beta}} - \cancel{\frac{\partial x'^\rho}{\partial x^\lambda}\partial_\rho\left(\frac{\partial x^\alpha}{\partial x'^\mu}\right)\frac{\partial x^\beta}{\partial x'^\nu}T_{\alpha\beta}} - \cancel{\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x'^\rho}{\partial x'^\lambda}\partial_\rho\left(\frac{\partial x'^\beta}{\partial x^\nu}\right)T_{\alpha\beta}} = 0$$

□

$\nabla_\rho T_{\alpha\beta}$

$$\nabla'_\lambda T'_{\mu\nu} = \partial'_\lambda T'_{\mu\nu} - \Gamma'^\sigma_{\lambda\mu}T'_{\sigma\nu} - \Gamma'^\sigma_{\lambda\nu}T'_{\mu\sigma} = \frac{\partial x^\rho}{\partial x'^\lambda}\frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\left[\partial_\rho\left(T_{\alpha\beta}\right) - \Gamma^\ell_{\rho\alpha}T_{\ell\beta} - \Gamma^\ell_{\rho\beta}T_{\alpha\ell}\right]$$

7. La métrica del espacio-tiempo de Friedmann-Robertson-Walker (FRW) en coordenadas cartesianas está dada por

$$ds^2 = c^2dt^2 - a(t)^2(dx^2 + dy^2 + dz^2),$$

donde la función $a(t)$ se conoce como *factor de escala*. Calcule todas las componentes no nulas de la conexión del espacio-tiempo de FRW en estas coordenadas.

$$g_{\mu\nu} = \begin{pmatrix} c^2 & 0 & 0 & 0 \\ 0 & -a(t)^2 & 0 & 0 \\ 0 & 0 & -a(t)^2 & 0 \\ 0 & 0 & 0 & -a(t)^2 \end{pmatrix} \quad ; \quad g^{\mu\nu} = \begin{pmatrix} 1/c^2 & 0 & 0 & 0 \\ 0 & -1/a(t)^2 & 0 & 0 \\ 0 & 0 & -1/a(t)^2 & 0 \\ 0 & 0 & 0 & -1/a(t)^2 \end{pmatrix}$$

Simbolos de Christoffel:

$$\Gamma^{\lambda}_{\mu\nu} = \tfrac{1}{2} g^{\lambda\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$$

Para $\lambda = 0$ (solo $\neq 0$, con $\sigma = 0$)

$$\Gamma^0_{\mu\nu} = \tfrac{1}{2} g^{0\sigma}(\dots)$$

$$\Gamma^0_{\mu\nu} = \tfrac{1}{2} g^{00}(\partial_\mu g_{0\nu} + \partial_\nu g_{0\mu} - \partial_0 g_{\mu\nu})$$

$\partial_\mu = \frac{\partial}{\partial x^\mu}$

$\mu = 0, 1, 2, 3$

$\frac{\partial}{\partial x^0} = \frac{\partial}{\partial t}$

$\frac{\partial}{\partial x^1} = \frac{\partial}{\partial x}$

$\frac{\partial}{\partial x^2} = \frac{\partial}{\partial y}$

$\frac{\partial}{\partial x^3} = \frac{\partial}{\partial z}$

$$\Gamma^0_{00} = \tfrac{1}{2} g^{00}(\partial_0 g_{00} + \partial_0 g_{00} - \partial_0 g_{00})$$

$$\Gamma^0_{00} = \tfrac{1}{2} g^{00}\left(\frac{\partial}{\partial t}(c^2)\right) = 0$$

$$\Gamma^0_{ij} = \tfrac{1}{2} g^{00}(\partial_i g_{0j} + \partial_j g_{0i} - \partial_0 g_{ij}) \quad (\partial_i g_{0j} = 0,\ \partial_j g_{0i} = 0)$$

$$\Gamma^0_{ij} = -\tfrac{1}{2} g^{00} \cdot \partial_0 g_{ij}$$ solo términos en ij

$$\Gamma^0_{ij} = -\frac{1}{2} \cdot \frac{1}{c^2} \cdot \frac{\partial}{\partial t}(-a(t)^2)\ \delta_{ij}$$

$$\Gamma^0_{ii} = \frac{1}{2c^2} \cdot 2a(t)\dot{a}(t) = \frac{a\dot{a}}{c}$$

$$\Gamma^0_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{a\dot{a}}{c} & 0 & 0 \\ 0 & 0 & \frac{a\dot{a}}{c} & 0 \\ 0 & 0 & 0 & \frac{a\dot{a}}{c} \end{pmatrix}$$

Para $\lambda = 1$ (solo $\neq 0$, con $\sigma = 1$)

$$\Gamma^1_{\mu\nu} = \tfrac{1}{2} g^{1\sigma}(\dots)$$

$$\Gamma^1_{\mu\nu} = \tfrac{1}{2} g^{11}(\partial_\mu g_{1\nu} + \partial_\nu g_{1\mu} - \partial_1 g_{\mu\nu}) \quad (\partial_1 g_{\mu\nu} = 0)$$

$$\Gamma^1_{00} = \tfrac{1}{2} g^{11}(\partial_0 g_{10} + \partial_0 g_{10}) = 0 \quad (g_{10} = 0)$$

$$\Gamma^1_{ij} = \tfrac{1}{2} g^{11} \left(\partial_i g_{1j} + \partial_j g_{1i} - \cancel{\partial_1 g_{ij}}^{\,0} \right)$$

$$\Gamma^1_{01} = \tfrac{1}{2} g^{11} \left(\partial_0 g_{11} + \cancel{\partial_1 g_{10}}^{\,0} \right) = \tfrac{1}{2} \left(-\tfrac{1}{a(t)^2} \right) \cdot -2\, a(t)\, \dot{a}(t) = \dot{a}(t)/a(t)$$

$$\Gamma^1_{10} = \tfrac{1}{2} g^{11} \left(\cancel{\partial_1 g_{10}}^{\,0} + \partial_0 g_{11} \right) = \tfrac{1}{2} \left(-\tfrac{1}{a(t)^2} \right) \cdot -2\, a(t)\, \dot{a}(t) = \dot{a}(t)/a(t)$$

$$\Gamma^1_{\mu\nu} = \begin{pmatrix} 0 & \frac{\dot{a}(t)}{a(t)} & 0 & 0 \\ \frac{\dot{a}(t)}{a(t)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Para $\lambda = 2$

$\neq 0$ solo en $\sigma = 2$

$$\Gamma^2_{\mu\nu} = \tfrac{1}{2} g^{2\sigma} \left(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu} \right)$$

$$\Gamma^2_{\mu\nu} = \tfrac{1}{2} g^{22} \left(\partial_\mu g_{2\nu} + \partial_\nu g_{2\mu} - \cancel{\partial_2 g_{\mu\nu}}^{\,0} \right)$$

con $\mu = 0$, $\nu = 2$
∧ $\mu = 2$, $\nu = 0$

$$\Gamma^2_{02} = \tfrac{1}{2} \left(-\frac{1}{a(t)^2} \right) \left(\frac{\partial}{\partial t} (-a(t)^2 + \cancel{\partial_2 g_{20}}^{\,0} \right) = \dot{a}(t)/a(t)$$

$$\Gamma^2_{20} = \tfrac{1}{2} \left(-\frac{1}{a(t)^2} \right) \left(\cancel{\partial_2 g_{20}}^{\,0} + \frac{\partial}{\partial t} g_{22} \right) = \dot{a}(t)/a(t)$$

$$\Gamma^2_{\mu\nu} = \begin{pmatrix} 0 & 0 & \frac{\dot{a}(t)}{a(t)} & 0 \\ 0 & 0 & 0 & 0 \\ \frac{\dot{a}(t)}{a(t)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Para $\lambda = 3$

$\neq 0$ solo en $\sigma = 3$

$$\Gamma^3_{\mu\nu} = \tfrac{1}{2} g^{3\sigma} \left(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu} \right)$$

$$\Gamma^3_{\mu\nu} = \tfrac{1}{2} g^{33} \left(\partial_\mu g_{3\nu} + \partial_\nu g_{3\mu} - \cancel{\partial_3 g_{\mu\nu}}^{\,0} \right)$$

con $\mu = 0$ $\nu = 3$ $\quad \partial_0 g_{33}$
∧ $\mu = 3$, $\nu = 0$

$$\Gamma^3_{03} = \tfrac{1}{2} \left(\frac{-1}{a(t)^2} \right) \cdot -2\, a(t) (\dot{a}(t)) = \frac{\dot{a}(t)}{a(t)}$$

$$\Gamma^3_{30} = -\tfrac{1}{2} \left(\frac{1}{a(t)^2} \right) \cdot -2 a(t) (\dot{a}(t)) = \frac{\dot{a}(t)}{a(t)}$$

$$\Gamma^3_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & \frac{\dot{a}(t)}{a(t)} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{\dot{a}(t)}{a(t)} & 0 & 0 & 0 \end{pmatrix}$$

8. La métrica de Schwarzschild en coordenadas esféricas está dada por

$$ds^2 = A(r)c^2dt^2 - B(r)dr^2 - r^2(d\theta^2 + \sin^2\theta d\phi^2),$$

donde $A(r)$ y $B(r)$ son funciones que dependen solo de la coordenada r y que se mostrarán explícitamente en las últimas semanas del curso. Calcule todas las componentes no nulas de la conexión de dicha métrica.

$$g_{\mu\nu} = \begin{pmatrix} A(r)c^2 & 0 & 0 & 0 \\ 0 & -B(r) & 0 & 0 \\ 0 & 0 & -r^2 & 0 \\ 0 & 0 & 0 & -r^2\sin^2\theta \end{pmatrix} \quad ; \quad g^{\mu\nu} = \begin{pmatrix} \frac{1}{A(r)c^2} & 0 & 0 & 0 \\ 0 & -\frac{1}{B(r)} & 0 & 0 \\ 0 & 0 & -\frac{1}{r^2} & 0 \\ 0 & 0 & 0 & -\frac{1}{r^2\sin^2\theta} \end{pmatrix}$$

Simbolos de Christoffel:

$$\Gamma^{\lambda}_{\mu\nu} = \tfrac{1}{2} g^{\lambda\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$$

Para $\lambda = 0$

$$\Gamma^0_{\mu\nu} = \tfrac{1}{2} g^{0\sigma}(\dots) \quad \text{solo} \neq 0, \text{ con } \sigma = 0$$

$$\Gamma^0_{\mu\nu} = \tfrac{1}{2} g^{00}(\partial_\mu g_{0\nu} + \partial_\nu g_{0\mu} - \partial_0 g_{\mu\nu})$$

$\partial_\mu = \frac{\partial}{\partial x^\mu}$

$\mu = 0, 1, 2, 3$

$\frac{\partial}{\partial x^0} = \frac{\partial}{\partial t}$

$\frac{\partial}{\partial x^1} = \frac{\partial}{\partial r}$

$\frac{\partial}{\partial x^2} = \frac{\partial}{\partial \theta}$

$\frac{\partial}{\partial x^3} = \frac{\partial}{\partial \varphi}$

$$\Gamma^0_{00} = \tfrac{1}{2} g^{00}(\partial_0 g_{00} + \partial_0 g_{00} - \partial_0 g_{00}) = 0$$

$$\Gamma^0_{00} = \tfrac{1}{2} g^{00}\left(\frac{\partial}{\partial t}(A(r)c^2)\right) = 0$$

$$\Gamma^0_{ij} = \tfrac{1}{2} g^{00}(\partial_i g_{0j} + \partial_j g_{0i} - \partial_0 g_{ij})$$

$i = 1, j = 0$

$i = 0, j = 1$

$$\Gamma^0_{10} = \tfrac{1}{2}\frac{1}{A(r)c^2}(\partial_1 g_{00} + \partial_0 g_{01}) = \frac{1}{2A(r)c^2}\cdot \dot{A}(r)c^2 = \frac{\dot{A}}{2A}$$

$$\Gamma^0_{01} = \tfrac{1}{2}\frac{1}{A(r)c^2}(\partial_0 g_{01} + \partial_1 g_{00}) = \frac{\dot{A}}{2A}$$

$$\Gamma^0_{\mu\nu} = \begin{pmatrix} 0 & \dot{A}/2A & 0 & 0 \\ \dot{A}/2A & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

para $\lambda = 1$:

$$\Gamma^1_{\mu\nu} = \tfrac{1}{2} g^{11}(\partial_\mu g_{1\nu} + \partial_\nu g_{1\mu} - \partial_1 g_{\mu\nu})$$

$$\Gamma^1_{\mu\nu} = \tfrac{1}{2} g^{11}(\partial_0 g_{10} + \partial_0 g_{10} - \partial_1 g_{00}) = 0$$

$\nu = 0, \mu = 0 \Rightarrow -\dot{A}c^2$

$$\Gamma^1_{00} = \tfrac{1}{2}\cdot -\frac{1}{B(r)}\cdot -\dot{A}c^2 = \frac{\dot{A}}{2B}c^2$$

$$\Gamma^1_{ij} = \tfrac{1}{2} g^{11}(\partial_i g_{1j} + \partial_j g_{1i} - \partial_1 g_{ij})$$

$i = 1, j = 1$

$$\Gamma^1_{11} = \tfrac{1}{2} g^{11}(\partial_1 g_{11}) = \tfrac{1}{2}\left(-\frac{1}{B(r)}\right)\cdot\frac{\partial}{\partial r}(-B(r)) = \frac{1}{2}\frac{\dot{B}(r)}{B(r)}$$

$$\Gamma^1_{ij} = \tfrac{1}{2} g^{11} (\partial_i g_{1j} + \partial_j g_{1i} - \partial_1 g_{ij})$$

$i=2, j=2$

$$\Gamma^1_{22} = \tfrac{1}{2} g^{11} (-\partial_1 g_{22}) \qquad -\tfrac{1}{2}\frac{\partial}{\partial r}\cdot r^2$$

$$= \tfrac{1}{2}\left(\frac{1}{B(r)}\right)(-2r) = \frac{-r}{B(r)}$$

con $i=3, j=3$

$$\Gamma^1_{33} = \tfrac{1}{2} g^{11} (-\partial_1 g_{33}) \qquad \partial/r\,(-r^2 \operatorname{sen}^2\theta)$$

$$\Gamma^1_{33} = -\tfrac{1}{2}\left(\frac{1}{B(r)}\right)(-2r\operatorname{sen}^2\theta) \qquad -r^2 \to -2r^2$$

$$\Gamma^1_{33} = \frac{-r\operatorname{sen}^2\theta}{B(r)}$$

$$\Gamma^1_{\mu\nu} = \begin{pmatrix} \frac{\dot{A}}{2B}c^2 & 0 & 0 & 0 \\ 0 & \frac{\dot{B}}{2B} & 0 & 0 \\ 0 & 0 & -r/B & 0 \\ 0 & 0 & 0 & \frac{-r\operatorname{sen}^2\theta}{B} \end{pmatrix}$$

Con $\lambda = 2$: $\quad \sigma = 2$

$$\Gamma^2_{\mu\nu} = \tfrac{1}{2} g^{22} (\partial_\mu g_{2\nu} + \partial_\nu g_{2\mu} - \partial_2 g_{\mu\nu})$$

para $\nu=2, \mu=1$

$$\Gamma^2_{12} = \tfrac{1}{2}(-1/r^2)(\partial_1 g_{22}) = 1/r$$

$$\frac{\partial}{\partial r}(-r^2) \qquad 2r$$

$\nu=1, \mu=2$

$$\Gamma^2_{21} = \tfrac{1}{2}(-1/r^2)(\partial_2 g_{21} + \partial_1(-r^2)) = 1/r$$

para $\nu = \mu = 3$

$$\Gamma^2_{\mu\nu} = \tfrac{1}{2} g^{22} (\partial_3 g_{23} + \partial_3 g_{23} - \partial_2 g_{33})$$

$$= \tfrac{1}{2}\left(-\frac{1}{r^2}\right)\left[-\frac{\partial}{\partial\theta}(-r^2\operatorname{sen}^2\theta)\right]$$

$$= -1/2 \cdot 2\operatorname{sen}\theta\cos\theta = -\operatorname{sen}\theta\cos\theta$$

$$\Gamma^2_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1/r & 0 \\ 0 & 1/r & 0 & 0 \\ 0 & 0 & 0 & -\operatorname{sen}\theta\cos\theta \end{pmatrix}$$

con $\lambda = 3$

$$\Gamma^{3}_{\mu\nu} = \tfrac{1}{2} g^{3\overset{\sigma=3}{3}} \left(\partial_\mu g_{3\nu} + \partial_\nu g_{3\mu} - \cancel{\partial_3 g_{\mu\nu}}^{\,0}\right)$$

$\nu = 3, \quad \mu = 1$

$\mu = 1, \quad \nu = 3$

$$\Gamma^{3}_{13} = \cancel{\tfrac{1}{2}} \left(\frac{1}{r^2 \operatorname{sen}^2\theta}\right)(\partial_1 g_{33})$$

$$\frac{\partial}{\partial r}\left(-r^2 \operatorname{sen}^2\theta\right) = -2r \operatorname{sen}^2\theta$$

$$\Gamma^{3}_{13} = \Gamma^{3}_{31} = 1/r$$

$\nu = 3, \quad \mu = 2$

$\mu = 2, \quad \nu = 3$

$$\Gamma^{3}_{\mu\nu} = \tfrac{1}{2} g^{33} \overset{\sigma=3}{\left(\partial_\mu g_{3\nu} + \partial_\nu g_{3\mu} - \cancel{\partial_3 g_{\mu\nu}}^{\,0}\right)}$$

$$\Gamma^{3}_{23} = \cancel{\tfrac{1}{2}} \left(-\frac{1}{r^2 \operatorname{sen}^2\theta}\right)(\partial_2 g_{33})$$

$$\frac{1}{\partial\theta}\left(-r^2 \operatorname{sen}^2\theta\right) \to -r^2 \cdot 2 \operatorname{sen}\theta \cos\theta$$

$$\Gamma^{3}_{23} = \frac{\cos\theta}{\operatorname{sen}\theta} = \cot(\theta)$$

$$\Gamma^{3}_{32} = \cot(\theta)$$

$$\Gamma^{3}_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/r \\ 0 & 0 & 0 & \cot\theta \\ 0 & 1/r & \cot\theta & 0 \end{pmatrix}$$

9. La métrica de Schwarzschild en coordenadas esféricas está dada por

$$ds^2 = \left(1 - \frac{r_s}{r}\right) c^2 dt^2 - \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 - r^2(d\theta^2 + \sin^2\theta d\phi^2),$$

donde $r_s = \dfrac{2GM}{c^2}$ se conoce como el *radio de Schwarzschild*. Calcule todas las componentes no nulas de la conexión de dicha métrica.

$$\Gamma^0_{\mu\nu} = \begin{pmatrix} 0 & \dot{A}/2A & 0 & 0 \\ \dot{A}/2A & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

↳ reemplazando:

con $A = \left(1 - \frac{r_s}{r}\right)c^2$, $B = \dfrac{1}{\left(1 - \frac{r_s}{r}\right)}$

$A = \dfrac{(r - r_s)c^2}{r}$ $\quad B = \dfrac{r}{r - r_s}$

$\dot{A} = \dfrac{\partial A}{\partial r} = + \dfrac{r_s}{r^2} c^2$

$\Rightarrow \dfrac{\dot{A}}{2A} = \dfrac{r_s \; r \cdot c^2}{2r^2 (r - r_s) c^2} = \dfrac{r_s}{2 \cdot r(r - r_s)}$

$$\Gamma^0_{\mu\nu} = \begin{pmatrix} 0 & \frac{r_s}{2r(r-r_s)} & 0 & 0 \\ \frac{r_s}{2r(r-r_s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Gamma^1_{\mu\nu} = \begin{pmatrix} \frac{\dot{A}}{2B}c^2 & 0 & 0 & 0 \\ 0 & \frac{\dot{B}}{2B} & 0 & 0 \\ 0 & 0 & -r/B & 0 \\ 0 & 0 & 0 & \frac{-r\operatorname{sen}^2\theta}{B} \end{pmatrix}$$

↳ reemplazando:

↳ $\dfrac{\dot{A}}{2B} c^2 = \dfrac{r_s}{2r^2} \cdot c^2 \cdot \left(\dfrac{r - r_s}{r}\right) = \dfrac{c^2 r_s (r - r_s)}{2r^3}$

↳ $\dot{B} = \dfrac{\partial}{\partial r}\left(\dfrac{r}{r - r_s}\right) = \dfrac{(1)(r - r_s) - r(1)}{(r - r_s)^2}$

↳ $\dfrac{\dot{B}}{2B} = \dfrac{-r_s}{(r - r_s)^2} \cdot \dfrac{(r - r_s)}{r} = \dfrac{-r_s}{2(r - r_s)r}$

↳ $r/B = r \cdot \dfrac{(r - r_s)}{r} = \dfrac{(r - r_s)}{r}$

↳ $r \operatorname{sen}^2\theta \cdot \dfrac{(r - r_s)}{r} = (r - r_s)\operatorname{sen}^2\theta$

$$\Gamma^1_{\mu\nu} = \begin{pmatrix} \frac{c^2 r_s (r - r_s)}{2r^3} & 0 & 0 & 0 \\ 0 & \frac{-r_s}{2r(r - r_s)} & 0 & 0 \\ 0 & 0 & -(r - r_s) & 0 \\ 0 & 0 & 0 & -(r - r_s)\operatorname{sen}^2\theta \end{pmatrix}$$

y las demás no cambian:

$$\Gamma^2_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1/r & 0 \\ 0 & 1/r & 0 & 0 \\ 0 & 0 & 0 & -\operatorname{sen}\theta\cos\theta \end{pmatrix}$$

$$\Gamma^3_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/r \\ 0 & 0 & 0 & \cot\theta \\ 0 & 1/r & \cot\theta & 0 \end{pmatrix}$$